

A disproof of Conjecture 00000002167

The spectral-radius maximizer among graphs of girth ≥ 5 is not unique

1 The conjecture

Conjecture 00000002167 reads:

Definition: The spectral radius of n -vertex graphs of girth ≥ 5 . *Conjecture:* The spectral radius is attained by Moore-approximate graphs (Brown type or PC-lab type); the classification of extremal graphs holds for all $n \geq 5$ (uniqueness of spectral-radius extrema).

The statement is a conjunction. Its last clause, made explicit by the parenthetical, is:

(U) for every $n \geq 5$, the graphs that attain the maximum spectral radius among n -vertex graphs of girth ≥ 5 are unique up to isomorphism.

The terms “Moore-approximate”, “Brown type” and “PC-lab type” are not defined in the statement, and we do not need them: we show that (U) is false at $n = 5$, so the conjunction is false whatever the first clause means. The failure persists if only connected graphs are considered.

2 Definitions

Definition 1. The *girth* of a graph is the length of a shortest cycle; a graph with no cycle (a forest) has girth ∞ . “If the graph does not contain any cycles (that is, it is a forest), its girth is defined to be infinity” (Wikipedia, *Girth (graph theory)*, [https://en.wikipedia.org/wiki/Girth_\(graph_theory\)](https://en.wikipedia.org/wiki/Girth_(graph_theory))). In Lean this is Mathlib’s `SimpleGraph.egirth` (value in $\mathbb{N} \cup \{\infty\}$, equal to $\top$ exactly for acyclic graphs), and “girth ≥ 5 ” is `5 <= G.egirth`.

Definition 2. The *spectral radius* $\rho(G)$ of a finite graph is $\max\{|\lambda| : \lambda \text{ an eigenvalue of its adjacency matrix}\}$. In Lean, `adjSpectralRadius G` is Mathlib’s `spectralRadius C (G.adjMatrix C)`, i.e. the supremum of $\|\lambda\|$ over the spectrum of the complex adjacency matrix.

C_5 is Mathlib’s `cycleGraph 5` (vertex i adjacent to $i \pm 1 \pmod 5$). $K_{1,4}$ is `starGraph 4` on $\{0, \dots, 4\}$: $i \sim j$ iff $i \neq j$ and $0 \in \{i, j\}$. C_5 is the smallest girth-5 (diameter-2) Moore graph: Wikipedia (*Moore graph*) lists “ C_5 with diameter 2, girth 5, degree 2, and order 5”.

3 The bound $\rho(G) \leq \sqrt{n-1}$ for girth ≥ 5

Lemma 3. Let G have girth ≥ 5 . Then G has no triangle $a \sim b \sim c \sim a$, and no 4-cycle $a \sim b \sim c \sim d \sim a$ with $a \neq c$, $b \neq d$. Conversely, a graph with neither has girth ≥ 5 .

Proof. A triangle or such a quadrilateral is a cycle of length 3 or 4. Conversely, every cycle has length ≥ 3 , and a cycle of length 3 or 4 is a triangle or such a quadrilateral (its support is duplicate-free). $\square$

Lemma 4. *If G has girth ≥ 5 and n vertices, then $\sum_{l \sim i} \deg l \leq n - 1$ for every vertex i .*

Proof. For $l \sim i$ let $T_l = N(l) \setminus \{i\}$, so $|T_l| = \deg l - 1$. If $j \in T_l \cap T_{l'}$ with $l \neq l'$, then $i \sim l \sim j \sim l' \sim i$ is a 4-cycle with $i \neq j$, $l \neq l'$. If $j \in T_l$ and $j \sim i$, then i, l, j is a triangle. So the T_l are pairwise disjoint subsets of $V \setminus (\{i\} \cup N(i))$, and $\sum_{l \sim i} (\deg l - 1) \leq n - 1 - \deg i$, i.e. $\sum_{l \sim i} \deg l \leq n - 1$. $\square$

Theorem 5. *If G has girth ≥ 5 and n vertices, every eigenvalue λ of its adjacency matrix A satisfies $|\lambda|^2 \leq n - 1$; hence $\rho(G) \leq \sqrt{n - 1}$.*

Proof. An element of the spectrum has an eigenvector $v \neq 0$ with $Av = \lambda v$ (the determinant of $\lambda I - A$ vanishes). Choose i with $|v_i|$ maximal; then $|v_i| > 0$. Since $(Av)_i = \sum_{j \sim i} v_j = \lambda v_i$,

$$|\lambda|^2 |v_i| = \left| \sum_{l \sim i} \sum_{j \sim l} v_j \right| \leq \sum_{l \sim i} \deg(l) |v_i| \leq (n - 1) |v_i|$$

by Lemma 4. Divide by $|v_i|$. $\square$

4 Two non-isomorphic extremal graphs on five vertices

Theorem 6 (main). (a) C_5 and $K_{1,4}$ have girth ≥ 5 , and both are connected.

(b) $\rho(C_5) = \rho(K_{1,4}) = 2$, and $\rho(G) \leq 2$ for every graph G of girth ≥ 5 on 5 vertices.

(c) Hence both C_5 and $K_{1,4}$ attain the maximum spectral radius among 5-vertex graphs of girth ≥ 5 , and also among connected ones.

(d) $C_5 \not\cong K_{1,4}$.

Proof. (a) Neither graph contains a triangle or a quadrilateral as in Lemma 3 (a finite check over $\{0, \dots, 4\}$). C_5 is connected (Mathlib `cycleGraph_connected`); every vertex of $K_{1,4}$ is 0 or adjacent to 0. (b) The all-ones vector is an eigenvector of C_5 with eigenvalue 2, since C_5 is 2-regular. For $K_{1,4}$, $v = (2, 1, 1, 1, 1)$ gives $Av = (4, 2, 2, 2, 2) = 2v$. So 2 lies in both spectra and $\rho \geq 2$; Theorem 5 with $n = 5$ gives $\rho \leq \sqrt{4} = 2$ for every girth- ≥ 5 graph on 5 vertices. (c) follows from (a) and (b). (d) An isomorphism preserves the number of edges, and C_5 has 5 edges while $K_{1,4}$ has 4. $\square$

Corollary 7. *Clause (U) fails at $n = 5$, both among all graphs and among connected graphs. Hence Conjecture 00000002167 is false.*

Remark 8. The same happens at every order of a girth-5 Moore graph: $n = 10$ (Petersen graph and $K_{1,9}$) and $n = 50$ (Hoffman–Singleton graph and $K_{1,49}$), since in each case both graphs have spectral radius $\sqrt{n - 1}$. Only $n = 5$ is formalized; one counterexample suffices for the “for all $n \geq 5$ ” claim.

Remark 9 (reading). The refutation uses the standard convention that forests have girth ∞ , so that $K_{1,4}$ has girth ≥ 5 . With that convention the star $K_{1,n-1}$ attains the bound $\sqrt{n - 1}$ for every n , so it is an extremal graph for every n .

5 Lean 4 formalization

The directory `lean/` is a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1); the source is `Conjecture2167/Basic.lean` (namespace `C2167`). It contains:

- `adjSpectralRadius`, `starGraph`, and the predicates `IsSpecExtremal P G` (girth ≥ 5 , side condition P , and maximal spectral radius among all graphs H on `Fin n` with girth ≥ 5 and $P H$) and `UniqueExtremal n P` (any two such maximizers are isomorphic, `Nonempty (G  $\simeq$ g H)`);
- `no_triangle`, `no_quad`, `five_le_egirth` (Lemma 3), `sum_degree_neighbors_le` (Lemma 4), `norm_sq_le_of_mem_spectrum` and `spectralRadius_le_sqrt` (Theorem 5, for any finite vertex type);
- `main` (Theorem 6): $\rho(C_5) = 2$, $\rho(K_{1,4}) = 2$, the bound on `Fin 5`, `IsSpecExtremal` for both graphs with $P = \text{True}$ and with $P = \text{Connected}$, and `IsEmpty (cycleGraph 5  $\simeq$ g starGraph 4)`;
- `conjecture_false`: $\neg(\forall n \geq 5, \text{UniqueExtremal } n \text{ True}) \wedge \neg(\forall n \geq 5, \text{UniqueExtremal } n \text{ Connected})$.

There is no sorry, no `native_decide` and no added axiom. `#print axioms` for `C2167.conjecture_false`, `C2167.main` and `C2167.spectralRadius_le_sqrt` reports only `propext`, `Classical.choice` and `Quot.sound`. The file also compiles with `-DwarningAsError=true`. To check:

```
cd lean
lake exe cache get
lake build
```